

Course Outline for
Software Quality Engineering (Lab)

General Information

Course Number	SWE-
Credit Hours	2+1 (Theory Credit Hour = 2, Lab Credit Hours = 3)
Prerequisite	None
Course Coordinator	Not Specified

Course Objectives

This course provides an introduction to fundamentals of software quality engineering. It is aimed to explain SQE concepts, principles, practices and measures/metrics to develop the acquired knowledge and skills. This course offers an opportunity to students to apply learned software quality engineering concepts, principles, practices and measures/metrics in software development context through different activities. The course guides/train students to assess software quality based on the learned software quality engineering concepts, principles, practices and measures/metrics according to their skills

Assessment Criteria

S. No	Description	Marks(%)
1	Mid	10%
2	Assignments, Lab Tasks Submissions, Sessional	40%
3	Project	40%
4	Final Exam	10%

Course Learning Outcomes (CLOs)

No.	Course Learning Outcome	Domain	Level	Assessment Tool
CLO1-Comprehension	Upon completion of this course, students will understand a GitHub project creation, issues, defects management on	Cognitive	1,2	Lab Tasks

	GitHub repository			
C2- Application	By the end of this course, students will be capable of applying software quality engineering principles such as code reviews, peer programming reviews using Github repository project created earlier	Cognitive	3	Lab Tasks and Assignments
C3- Evaluation and Analysis	Upon completion of the course, students will possess the ability to critically evaluate software quality Using different techniques	Cognitive	4,5	Lab Tasks, Project

Domains: C=Cognitive, A=Affective, P=Psychomotor Levels:

Cognitive = {1: Remembering, 2: Understanding, 3: Applying, 4: Analyzing, 5: Evaluating, 5: Creating}

Affective = {1: Receiving, 2: Responding, 3: Valuing, 4: Organizing, 5: Characterizing}

Psychomotor= {1: Imitation, 2: Manipulation, 3: Precision, 4: Articulation, 5: Naturalization}

Below is a hands-on, industry-oriented Software Quality Engineering (SQE) lab task set built around GitHub. These labs progressively take students from repository setup and Git workflows to automated testing, code quality, CI/CD, defect management, and a final quality-engineering pipeline.

Course Contents (Week-wise Breakdown)

S. No	Task	Objective	CLO
Lab 1	GitHub Repository and SQE Project Setup	Learn how software-quality activities are integrated into a professional GitHub-based development workflow.	
Lab 2	Git Branching, Commits and Pull Requests	Practice professional Git branching and code-review workflows	
Lab 3	GitHub Issues as a Defect Management System	Learn professional defect reporting and tracking.	
Lab 4	Test Planning and Test Case Management	Develop a professional Software Test Plan and test cases.	
Lab 5	Black-Box Testing Using Equivalence Partitioning	Apply systematic test-design techniques.	
Lab 6	Boundary Value Analysis	Use Boundary Value Analysis to identify defects that occur at input boundaries.	
Lab 7	Unit Testing with GitHub	Develop automated unit tests.	
Mid Lab Exam			
Lab 8	Test-Driven Development with GitHub	Understand Test-Driven Development.	
Lab 9	Code Review Using GitHub Pull Requests	Perform professional code reviews.	
Lab 10	Static Code Analysis with SonarQube/SonarCloud	Detect code-quality problems automatically.	
Lab 11	Automated Test Coverage	Measure how much source code is exercised by automated tests.	
Lab 12	API Testing	Test REST APIs as an SQE engineer.	
Lab 13	Selenium-Based Automated UI Testing	Automate functional testing of a web application.	
Lab 14	Continuous Integration with GitHub Actions	Automatically execute quality checks whenever code changes	
Lab 15	Defect Lifecycle Management	Understand the complete software defect lifecycle.	
Final Lab Exam			